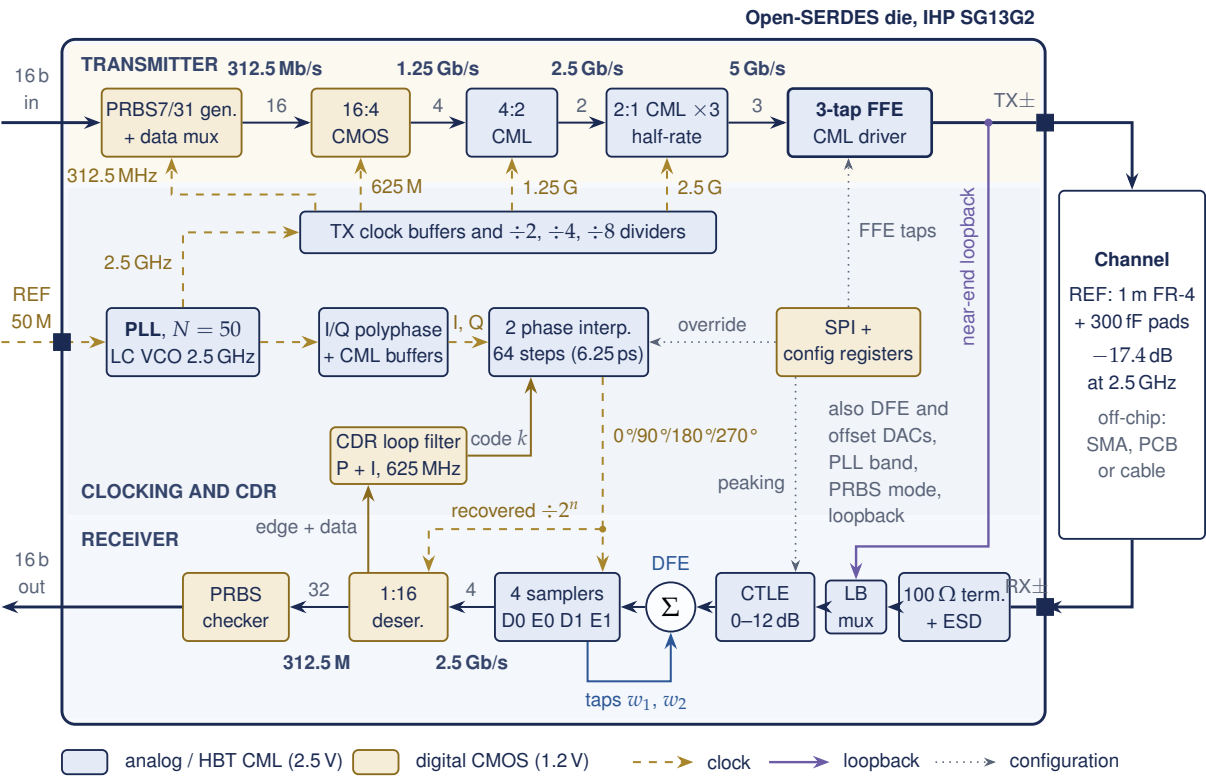

Open-SERDES die, IHP SG13G2
TRANSMITTER
312.5 Mb/s
1.25 Gb/s
2.5 Gb/s
5 Gb/s
16 b in
PRBS7/31 gen. + data mux
16
16:4 CMOS
4
4:2 CML
2
2:1 CML ×3 half-rate
3
3-tap FFE CML driver
TX±
312.5 MHz
625 M
1.25 G
2.5 G
TX clock buffers and ÷2, ÷4, ÷8 dividers
FFE taps
near-end loopback
2.5 GHz
REF 50 M
PLL, N = 50 LC VCO 2.5 GHz
I/Q polyphase + CML buffers
I, Q
2 phase interp. 64 steps (6.25 ps)
override
SPI + config registers
Channel
REF: 1 m FR-4 + 300 fF pads
−17.4 dB at 2.5 GHz
off-chip: SMA, PCB or cable
CDR loop filter P + I, 625 MHz
code k
0°/90°/180°/270°
also DFE and offset DACs, PLL band, PRBS mode, loopback
peaking
CLOCKING AND CDR
recovered ÷2^n
edge + data
RECEIVER
16 b out
DFE
PRBS checker
32
1:16 deser.
4
4 samplers D0 E0 D1 E1
Σ
CTLE 0–12 dB
LB mux
100 Ω term. + ESD
RX±
312.5 M
2.5 Gb/s
taps w1, w2
analog / HBT CML (2.5 V)
digital CMOS (1.2 V)
clock
loopback
configuration